

Chemical control of neckblast of rice

RAM SINGH

CCS Haryana Agricultural University Rice Research Station, Kaul-136 021
E-mail: singhddrram@gmail.com

Blast caused by *Pyricularia grisea* Sacc. [teleomorph: *Magnaporthe grisea* (Hebert) Barr.] is a devastating disease of rice which is known to cause substantial yield losses in different eco-systems of India (Laha *et al.*, 2016). In Haryana, it is a major hurdle in harvesting potential yield of export quality scented (*Basmati*) rice cultivars occupying about 50% of the total paddy acreage in the state (Anonymous, 2018). The disease appears as leaf blast, neck blast, node blast and collar rot at different stages of crop growth. Out of these, neck blast is the most destructive phase causing 1.8-37.5% reduction in grain yield depending upon the disease incidence (Singh *et al.*, 2013). The disease is also known to reduce milling quality and market value of rice. So far high level of resistance against neck blast has not been observed in commercially cultivated rice varieties. Hence, management of blast has primarily relied on foliar application of fungicides. Several fungicides have been found effective against neck blast (Singh *et al.*, 2017). However, continuous research is required to identify new effective fungicides for sustainable management of these diseases.

To evaluate the efficacy of fungicides against neck blast, field trials were conducted at CCS Haryana Agricultural University Rice Research Station, Kaul during *khari* 2017 and 2018 for neck blast in triplicate set of 5m x 2m plots in randomized block design by transplanting 30 days old seedlings of susceptible variety *Basmati* CSR30. Agronomic practices recommended for rice cultivation by CCSHAU Hisar were followed for raising the crop.

To evaluate the efficacy of fungicides against neck blast, eight combination fungicides *viz.* flusilazole 12.5% + carbendazim 25% (Lustre 37.5SE), azoxystrobin 18.2% + difenoconazole

11.4% (Amistar Top 325 SC), azoxystrobin 11% + tebuconazole 18.3% (Custodia 320 SC), tricyclazole 18% + mancozeb 62% WP (Merger 80WP), zineb 68% + hexaconazole 4% WP (Avtar 72 WP), trifloxystrobin 25% + tebuconazole 50% WG (Nativo 75 WG), mancozeb 50% + carbendazim 25% (Sprint 75WS) and fluxapyroxad 62.5 g/l + epoxiconazole 62.5 g/l EC (Adexar 125 EC) were sprayed at leaf blast initiation and 50% panicle emergence stage of the crop. Neck blast incidence was recorded from 25 randomly selected hills from each replicate of the treatments one week before harvest as described by Singh *et al.* (2010). Grain yield was recorded from individual plots and expressed as kg/ha.

The neck blast incidence in different treatments ranged from 5.52 to 14.94 and 4.18 to 18.94% during *khari* 2017 and 2018, respectively. All the fungitoxicants significantly reduced the disease incidence during both the years. Based on mean data, azoxystrobin 18.2% + difenoconazole 11.4 % SC at 1 ml/l, trifloxystrobin 25% + tebuconazole 50% WG at 0.4 g/l and azoxystrobin 11% + tebuconazole 18.3% SC (statistically at par) were found to be the most effective treatments which reduced the neck blast incidence by 71.3, 70.5 and 63.5% along with 13.2, 17.5 and 9.0% enhancement in grain yield of paddy, respectively. Zineb 68% + hexaconazole 4% WP was observed to be the least effective treatment (Table 1). Use of such a combination fungicides is safer from the point of view of fungicide resistance development in target pathogens due to its broad range of action.

Singh *et al.* (2010) have suggested the use of carpropamid and tricyclazole while Yamini Verma and Santha Kumari (2012) have reported the effectiveness of isprothiolane and carbendazim for blast management. Besides, tricyclazole and metominostrobin (Prasad *et al.*, 2011; Upmanyu and Rana, 2012), azoxystrobin, trifloxystrobin 25% + tebuconazole 50% and kresoxim methyl (Hubballi

Table 1. Effect of fungicides on neck blast incidence and grain yield of paddy variety *Basmati* CSR 30 during *kharif* 2017 and 2018

Treatments	Dosage/ litre	Neck blast incidence (%)			Grain yield (kg/ha)		
		2017	2018	Mean	2017	2018	Mean
flusilazole 12.5% + carbendazim 25% SE	1.0 ml	6.41 (14.63)	8.13 (16.50)	7.27 (15.56)	4367	4267	4317
azoxystrobin 18.2 % + difenoconazole 11.4 % SC	1.0 ml	5.56 (13.62)	4.18 (11.76)	4.87 (12.19)	4500	4333	4416
azoxystrobin 11% + tebuconazole 18.3% SC	1.5 ml	6.56 (14.83)	5.81 (13.89)	6.18 (14.36)	4267	4233	4250
tricyclazole 18 % + mancozeb 62 % WP	2.5 g	6.97 (15.96)	8.01 (16.39)	7.49 (16.17)	4467	4433	4450
zineb 68% + hexaconazole 4% WP	2.5 g	8.78 (17.21)	10.13 (18.51)	9.45 (17.86)	4233	4367	4300
trifloxystrobin 25% + tebuconazole 50% WG	0.4 g	5.52 (13.55)	4.57 (12.30)	4.99 (12.92)	4667	4500	4583
mancozeb 50% + carbendazim 25% WS	2.5 g	6.15 (14.35)	8.58 (17.01)	7.36 (15.68)	4433	4300	4367
fluxapyroxad 62.5 g/l + epoxiconazole 62.5 g/l EC	1.5 ml	6.45 (14.63)	8.06 (16.47)	7.25 (15.55)	4400	4267	4333
Control	-	14.94 (22.72)	18.94 (25.79)	16.94 (24.25)	3967	3833	3900
CD 5%		1.67	2.35	-	367	251	-
CV (%)		6.08	8.15	-	4.81	3.37	-

Figures in parentheses indicate angular transformed values

and Rabindran, 2014; Singh and Sunder, 2015; Kumar *et al.*, 2018), tricyclazole 22% + hexaconazole 3% (Magar *et al.*, 2015), tricyclazole 18% + mancozeb 62% WP and tricyclazole 45% + hexaconazole 10% WG (Singh *et al.*, 2017) have also been reported to be highly effective against this disease.

ACKNOWLEDGEMENTS

This work was funded by ICAR-Indian Institute of Rice Research, Hyderabad. The author is grateful to the Director, ICAR-Indian Institute of Rice Research, Hyderabad.

REFERENCES

- Anonymous** (2018). *Basmati Acreage and Yield Estimation in Punjab, Haryana, Delhi, Uttarakhand, Himachal Pradesh, Western Uttar Pradesh and Parts of Jammu & Kashmir (kharif-2018) Report -2*, September 18, 2018. Basmati Export Development Foundation (APEDA), New Delhi, 37 p.
- Hubballi, M. and Rabindran, R.** (2014). *J. Mycol. Pl. Pathol.* **44**: 46-53.
- Kumar, S., Kumari, S., Rai, S.N. and Sinha, A.K.** (2018). *Curr. J. Appl. Sci. Technol.* **31**: 1-9.
- Laha, G.S., Sailaja, B., Prasad, M.S., Ladhakshmi, D., Krishnaveni, D., Singh, R., Prakasam, V., Yugander, A., Kannan, C., Valarmathi, P. and Ravindra Babu, V.** (2016). *Changes in Rice Disease Scenario in India: An Analysis from Production Oriented Survey. Technical Bulletin No. 91*. ICAR-Indian Institute of Rice Research, Rajendranagar, Hyderabad, 95 p.
- Magar, P.B., Acharya, B. and Pandey, B.** (2015). *Intern. J. Appl. Sci. Biotechnol.* **3**: 474-478.
- Prasad, M.S., Sheshumadhav, M., Laha, G.S., Ladhakshmi, D., Krishnaveni, D., Mangrauthia, S.K., Balachandran, S.M., Sundaram, R.M., Arunakranthi, B., Madhanmohan, K., Ratnamadhavi, K., Kumar, V. and Viraktamath, B.C.** (2011). *Rice Blast Disease and its Management. Technical Bulletin No. 57*. Directorate of Rice Research (ICAR), Rajendranagar, Hyderabad, 52 p.
- Singh, R. and Sunder, S.** (2015). *J. Mycol. Pl. Pathol.* **45**: 55-59.
- Singh, R., Sunder, S. and Dodan, D.S.** (2013). *Indian Phytopath.* **66**: 249-251.
- Singh, R., Sunder, S., Dodan, D.S., Ram, L. and Singh, R.** (2010). *Indian Phytopath.* **63**: 212-215.
- Singh, R., Sunder, S. and Ram, M.** (2017). *Indian Phytopath.* **70**: 500-502.
- Upmanyu, S. and Rana, S.K.** (2012). *Pl. Dis. Res.* **27**: 92-93.
- Yamini Verma, C.K. and Santhakumari, P.** (2012). *Indian Phytopath.* **65**: 87-88.